

Utilizing LiDAR Data for 3D Sound Source Localization

Masahiko Goto
Yasuhiro Oikawa
Atsuto Inoue
Wataru Teraoka
gotoma@ruri.waseda.jp
yoikawa@waseda.jp
atsuto.inoue@aoni.waseda.jp
w.teraoka@kurenai.waseda.jp
Waseda University
Tokyo, Japan

Takahiro Sato
Yasuyuki Iwane
Masahito Kobayashi
Takahiro_Satou@tobishima.co.jp
Yasuyuki_Iwane@tobishima.co.jp
Masahito_Kobayashi@tobishima.co.jp
Tobishima Corporation
Chiba, Japan



Figure 1: (a) Scanning the observation space using an iOS device equipped with a LiDAR sensor. (b) shows the 3D data of the microphone array and the sound source. (c) Visualization of sound source estimation.

ABSTRACT

This paper introduces a visualization system of 3D sound pressure distribution using a minimum variance distortionless response (MVDR) beamformer with Light Detection and Ranging (LiDAR) technology to estimate the sound source localization. By using LiDAR to capture 3D data, the proposed system calculates the time-averaged output power of the MVDR beamformer at the virtual source position for each point in the point cloud data. The results are then superimposed onto the 3D data to estimate sound sources. The proposed system provides a more visually comprehensible display of the sound pressure distribution in 3D.

ACM Reference Format:

Masahiko Goto, Yasuhiro Oikawa, Atsuto Inoue, Wataru Teraoka, Takahiro Sato, Yasuyuki Iwane, and Masahito Kobayashi. 2023. Utilizing LiDAR Data for 3D Sound Source Localization. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '23 Posters)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588028.3603682>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Posters, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0152-8/23/08.

<https://doi.org/10.1145/3588028.3603682>

1 INTRODUCTION

Numerous studies have aimed to visualize the sound field to evaluate invisible sound information. Sound source localization is a technique used to identify not only the direct sound source but also the reflected sound source. However, most techniques are limited to identifying the actual location of the sound source in a 3D space because the 3D space information is not fully used and the estimated sound pressure distribution is displayed on a 2D cross-section. The proposed system utilizes minimum variance distortionless response (MVDR) beamformer [Johnson and Dudgeon 1993] to extend the 2D cross-sectional display of the sound pressure distribution to the entire 3D space. This paper proposes an investigating the use of 3D data with Light Detection and Ranging (LiDAR) [Weitkamp 2005] to make the sound pressure distribution display of a 3D sound source localization system more visually comprehensible. We also reports the results of that.

2 PROPOSED SYSTEM

An overview of the proposed system is presented in Fig. 2. The proposed system utilizes an iOS device equipped with LiDAR to acquire the 3D data of the observation space. A 12-channel planar microphone array was used for recording and was connected to a PC via an audio interface. The 3D data of the observation space, captured by the LiDAR scanner, is obtained as point cloud data. We assume that there is a virtual source at the position of each point of the point cloud data. The time-averaged output power of the MVDR beamformer is calculated at the virtual source position r . Figure 3

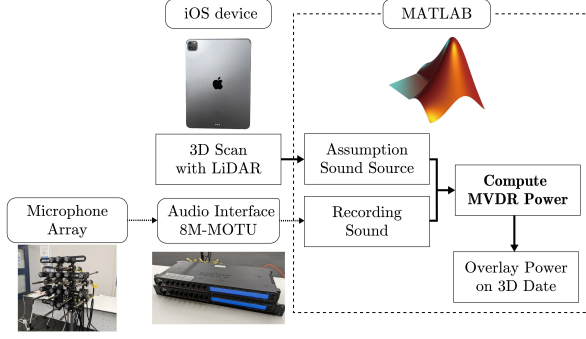


Figure 2: System Overview. The proposed system consists of a 12-channel microphone array, an audio interface, a PC and an iOS device. That uses 3D data of the observation space by using LiDAR and superimposes the estimated results on the 3D data to visualize the sound pressure distribution.

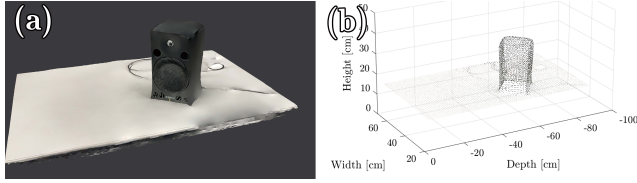


Figure 3: (a) 3D scan data using an iOS device. (b) Converting the 3D data into point cloud data for the beamformer.

shows the assumed sound source position, derived from the 3D data. The calculated results are then superimposed on the 3D data and displayed to estimate the sound source position. The setting of the virtual source position, the calculation of the beamformer output power, and the superimposition of the calculation results on the 3D data are performed in MATLAB.

2.1 MVDR beamformer

The proposed system employs MVDR beamformer, which applies a weight vector $\mathbf{w}$ by the observed signal vector $\mathbf{x}$ to obtain the output signal y , as $y = \mathbf{w}^H \mathbf{x}$, where H denotes Hermitian transpose. In the MVDR beamformer, weight vector $\mathbf{w}$ is determined as

$$\mathbf{w} = \frac{\mathbf{R}^{-1} \mathbf{a}(\mathbf{r})}{\mathbf{a}^H(\mathbf{r}) \mathbf{R}^{-1} \mathbf{a}(\mathbf{r})}, \quad (1)$$

where $\mathbf{R} = E[\mathbf{x}\mathbf{x}^H]$ is the signal correlation matrix. The output power of the MVDR beamformer is given by

$$P_{\text{MVDR}} = \frac{1}{\mathbf{a}^H(\mathbf{r}) \mathbf{R}^{-1} \mathbf{a}(\mathbf{r})}, \quad (2)$$

where $\mathbf{a}^H(\mathbf{r})$ is the complex conjugate of the sensor response vector.

2.2 Using 3D data with LiDAR

In recent years, LiDAR has gained attention as a means of capturing 3D information in space. Some Apple iOS devices are equipped with LiDAR sensors that are useful for measuring the shape of structures and for using point cloud 3D data.

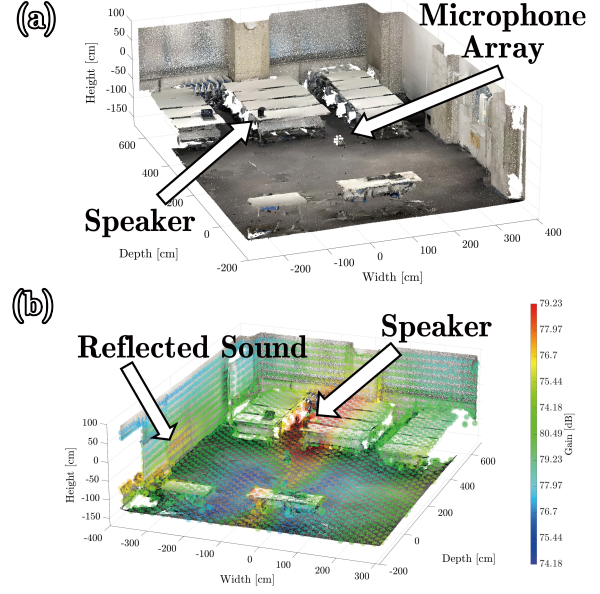


Figure 4: (a) 3D data of the observation space. (b) Visualization of the sound pressure distribution. The sound pressure level near the loudspeaker is 79.2 dB and that of the reflected sound is 76.8 dB.

3 VISUALIZATION EXPERIMENT

We conducted an experiment of visualization of sound pressure distribution using the proposed system. The experimental conditions included the following: a sound from a loudspeaker with white noise, an observed frequency range from 1,000 to 6,000 Hz. After setting up the experimental apparatus in the seminar room, the 3D data from the LiDAR scan using an iOS device is shown in Fig. 4 (a). This 3D data comprises 1,500,879 points. The loudspeaker was placed at the center of the room, and the microphone array was placed at the front side of the room. The sound was generated by the loudspeaker placing at an angle towards the left wall to verify the sound reflected from the wall. Fig. 4 (b) shows the sound pressure distribution on the room surface obtained by the proposed method. The MVDR beamformer output results for each of 3,249 virtual sources positions were superimposed onto the 3D data of the seminar room. Some parts with sound pressure level high can be observed close to the loudspeaker and on some parts of the left wall as shown in Fig. 4 (b). This allowed the confirmation of the direct sound from the loudspeaker and the sound reflected from the left wall. In the proposed system, it could be clear to visually understand the sound source localization on the surface of the room.

REFERENCES

- Don H. Johnson and Dan E. Dudgeon. 1993. *Array Signal Processing: Concepts and Techniques*. Prentice Hall PTR, Upper Saddle River, NJ.
- C. Weitkamp. 2005. *Lidar: range-resolved optical remote sensing of the atmosphere*. Springer, New York, NY.